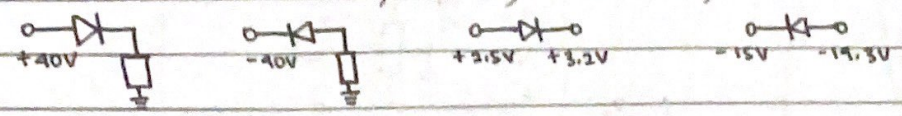
Hijriyani Nur Riska
1A - TNK
231344015

No.
Date:

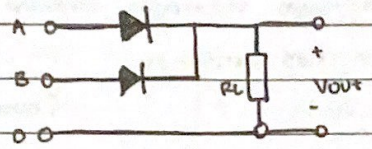
Soal Latihan Dioda

1 Tentukan keadaan masing - masing Dioda dengan kondisi bias DC di bawah ini

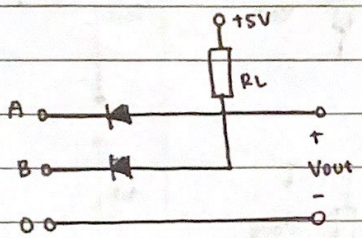


Kondisi dioda	Kondisi dioda	Kondisi dioda	Kondisi dioda
Forward			

2. Asumsikan dioda ideal. Tentukan logika keluaran tiap rangkaian dengan berbagai kemungkinan tegangan masukan, kemudian tentukan jenis gerbang logika

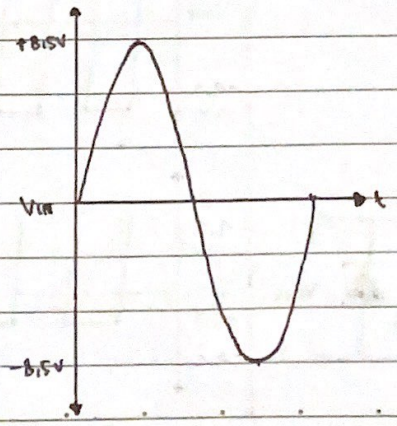
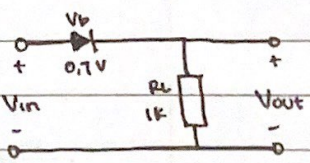
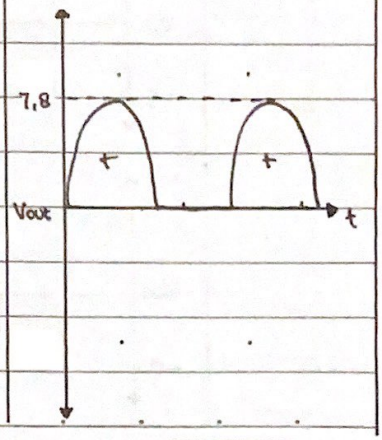


A	B	OUt	Nama gerbang logika
0	0	0	OR Gate
0	5	5	
5	0	5	
5	5	5	



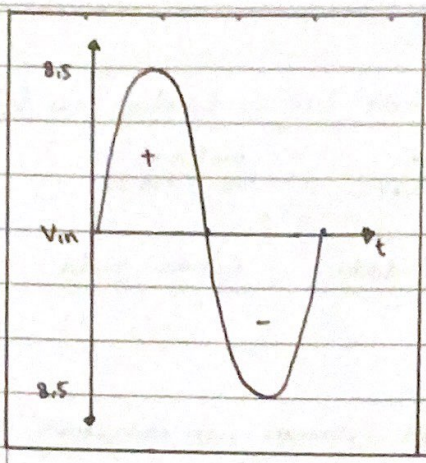
A	B	OUt	Nama gerbang logika
0	0	0	AND Gate
0	5	0	
5	0	0	
5	5	5	

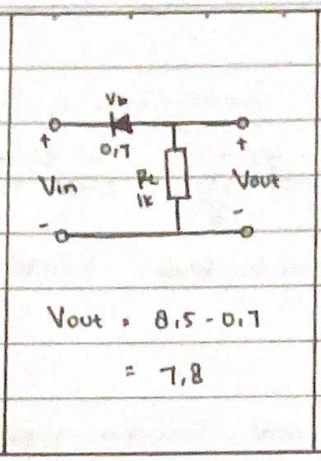
3 Gambarkan bentuk gelombang keluaran untuk masing - masing rangkaian :

Vinput	Rangkaian Dioda	Voutput
	a)  Vout = 8,5 - 0,7 = 7,8	

No.

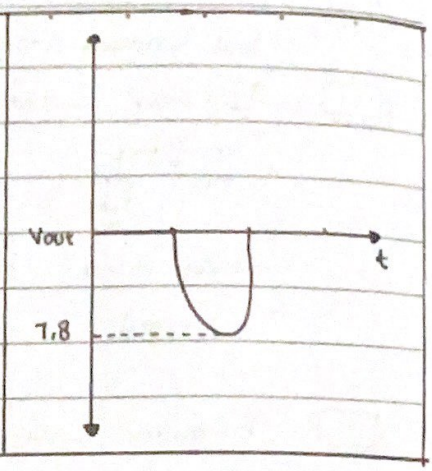
Date:



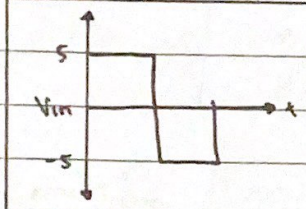
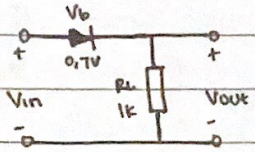
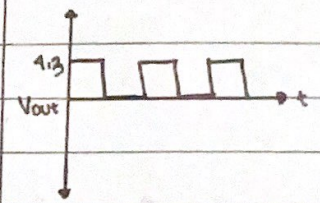
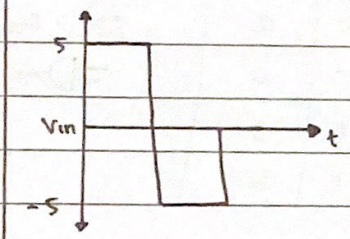
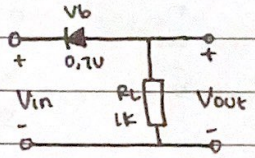
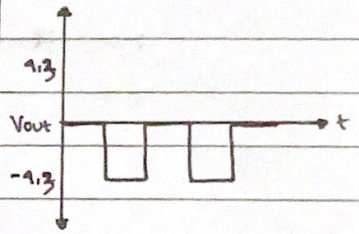
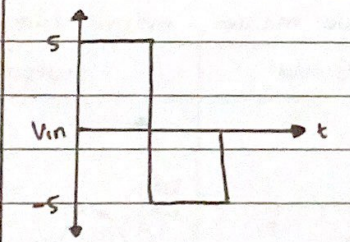
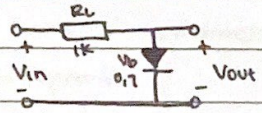
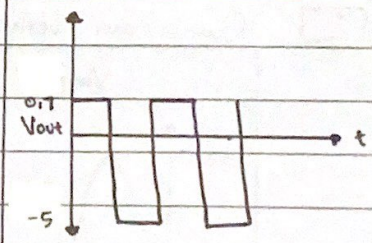
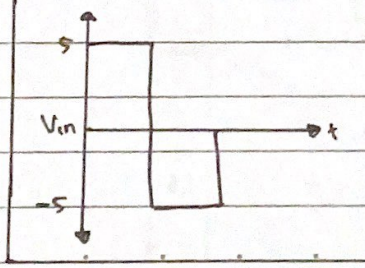
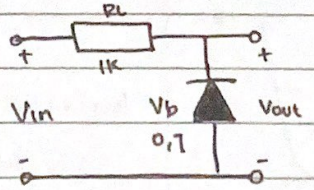
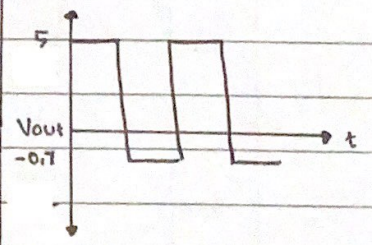


Vout = 8,5 - 0,7

= 7,8



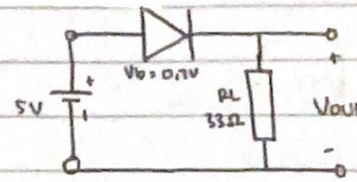
4. Asumsikan pendekatan dioda dengan mempunyai tegangan ambang $V_b = 0,7\ V$. Gambarkan bentuk gelombang keluaran tiap rangkaian

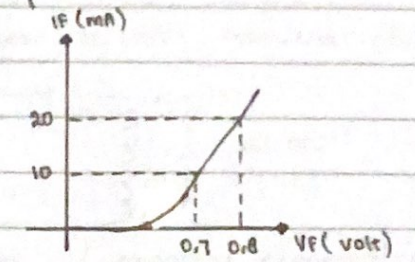
Vinput	Rangkaian Dioda	Voutput
		
		
		
		

No.

Date

5. Temukan besarnya IL dan tegangan pada RL bila :





a) Dioda ideal

$$I_L = \frac{5V}{33\Omega} = 0,1515\ A$$

$$V_L = 5V$$

b) Dioda tidak ideal dengan kurva karakteristik seperti pada Gambar.

$$V_L = 5 - 0,7$$

$$= 9,3\ V$$

$$I_L = \frac{V_{in} - V_b}{R_L} = \frac{5 - 0,7}{33} = \frac{9,3}{33} = 0,13\ A$$